LINEAR ATTENTION IS (MAYBE) ALL YOU NEED (TO UNDERSTAND TRANSFORMER OPTIMIZATION)

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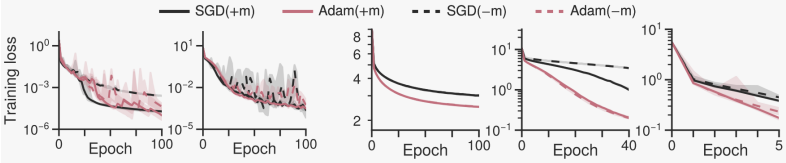
Introduction

- **Question:**
  - Why Transformer optimization is so difficult?
  - How do algorithms like **Adam perform better than SGD** in transformer training?



(a) **CNNs** on MNIST and CIFAR-10 (b) **Transformers** on PTB, WikiText2, and SQuAD

- **Major Obstacle:**
  - full-fledged transformers are extremely complicated to model

- **Main Contributions:**
  - propose **shallow linear transformer model** on linear regression
  - **reproduce all the optimization features** that have been reported for full transformers
    - Gap between SGD and Adam
    - Heavy-tailed stochastic noise
    - Robust condition number
    - Directional smoothness
  - How these features change when **changing settings** (data distribution, the number of layers)

Linear Transformer Architecture

- **Linear Regression Task**

  Input matrix:

$$X = \begin{bmatrix} x_1^1 & \cdots & x_1^n & x_1^{(n+1)} \\ \vdots & \ddots & \vdots & \vdots \\ x_d^1 & \cdots & x_d^n & x_d^{(n+1)} \\ y^1 & \cdots & y^n & 0 \end{bmatrix} \in \mathbb{R}^{(d+1)\times(n+1)}$$

- **Self-Attention of Transformer**

  $W_Q, W_K, W_V \in \mathbb{R}^{(n+1)\times k}$

  $Q, K, V = XW \in \mathbb{R}^{(d+1)\times k}$

$$\begin{aligned} \text{Attention} &= \text{softmax}\left(\frac{QK^\top}{\sqrt{k}}\right) V \\ &= \text{softmax}\left(\frac{XW_Q W_K^\top X^\top}{\sqrt{k}}\right) XW_V \end{aligned}$$

- **Linear Transformer**

$$\text{Attention} = PXM\left(X^\top QX\right)$$

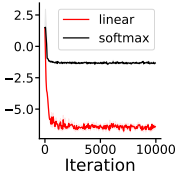
  where $P \sim W_V$, $Q \sim W_Q W_K^\top \in \mathbb{R}^{(d+1)\times(d+1)}$, and

$$M \equiv \begin{bmatrix} \mathbb{I}_n & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{(n+1)\times(n+1)}$$

- **Multi-Layer Linear Transformer**

  The output of the L-th layer attention, X_L, be recursively defined as

$$X_{\ell+1} = X_\ell + \frac{1}{n}\text{Attention}(X_\ell), \quad \ell = 0, 1, \cdots L-1$$



Why remove softmax?

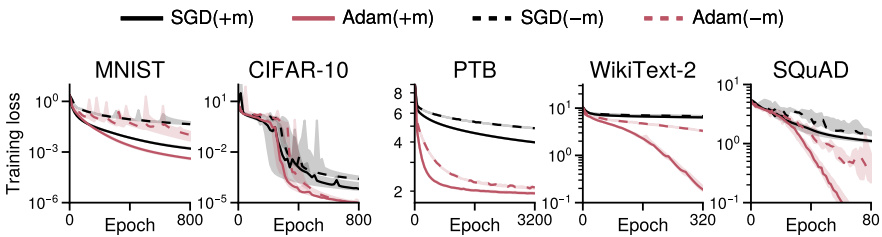
The linear attention unit is much better suited to the task oflinear regression.

(von Oswald et al., 2023)

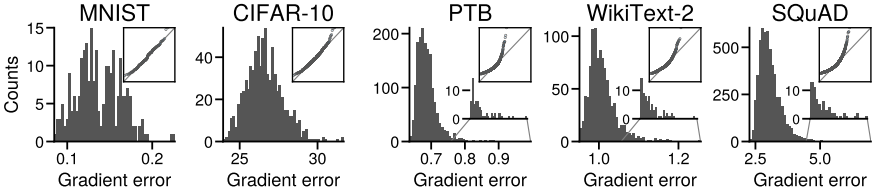
Distinctive Features of Transformer Optimization

- **Gap between SGD and Adam.**

Adaptive method like **Adam are significantly better than SGD**!



- **Heavy-tailed stochastic noise.** $||g(x) - \nabla f(x)||^2$



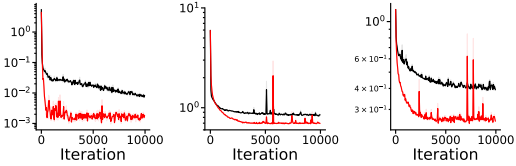
Quantile Quantile Plot

Distinctive Features of Transformer Optimization

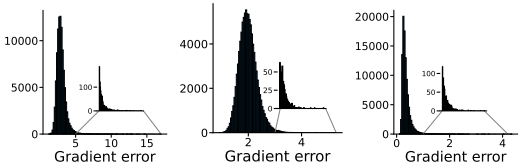
- **Reproduce in Linear Transformer**

$L = 3, d = 5$ is the minimally complex setting that still recovers all of the features of full transformers.

	Setting 1	**Setting 2**	**Setting 3**
#contexts n	20	5	20
distribution of x	$\mathcal{N}(0, I_d)$	$\mathcal{N}(0, I_d)$	$\sqrt{\Gamma_{0.1,10}} \cdot \mathrm{Unif}(\mathbb{S}^{d-1})$



Adam > SGD



Heavy-tailed

Distinctive Features of Transformer Optimization

- **Robust condition number**

  The condition number

$$\kappa_f(x) = \lambda_{\max}\left[\nabla^2 f(x)\right] / \lambda_{\min}\left[\nabla^2 f(x)\right]$$

  provided that $\lambda_{\min}\left[\nabla^2 f(x)\right] > 0$

  $\lambda_{\max}[M]$ and $\lambda_{\min}[M]$ are the largest and smallest eigenvalue of the symmetric matrix M

  (Jiang et al., 2022) consider a **robust** version of condition number defined as

$$R_{med}^{OPT} = \lambda_{\max}\left[\nabla^2 f\right] / \lambda_{\text{median}}\left[\nabla^2 f\right]$$

Layer#	**Iteration 750** $R_{\text{med}}^{\text{SGD}} / R_{\text{med}}^{\text{Adam}}$	**Iteration 1250** $R_{\text{med}}^{\text{SGD}} / R_{\text{med}}^{\text{Adam}}$
15	1.65 (0.65)	2.01 (1.00)
17	1.91 (0.53)	1.43 (0.63)
22	3.54 (1.21)	2.28 (1.18)

Transformers

	Iteration 750 $R_{\text{med}}^{\text{SGD}} / R_{\text{med}}^{\text{Adam}}$	**Iteration 1250** $R_{\text{med}}^{\text{SGD}} / R_{\text{med}}^{\text{Adam}}$
Setting 1	1.76 (0.40)	1.58 (0.41)
Setting 2	3.14 (0.97)	5.98 (2.86)
Setting 3	9.57 (13.3)	6.53 (3.55)

Linear Transformers

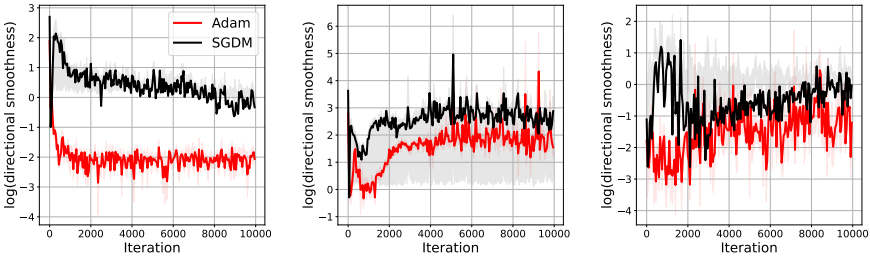
Distinctive Features of Transformer Optimization

- **Directional smoothness**

  Consider the following Taylor expansion along the iterates:

$$f(x_{t+1}) - f(x_t) = \nabla f(x_t)^\top (x_{t+1} - x_t) + \frac{1}{2}(x_{t+1} - x_t)^\top \nabla^2 f(x_t)(x_{t+1} - x_t) + \mathcal{O}(\eta^3)$$

  Define directional smoothness $= \dfrac{(x_{t+1} - x_t)^\top \nabla^2 f(x_t)(x_{t+1} - x_t)}{||x_{t+1} - x_t||^2}$

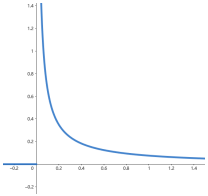


- **Spherical covariates**

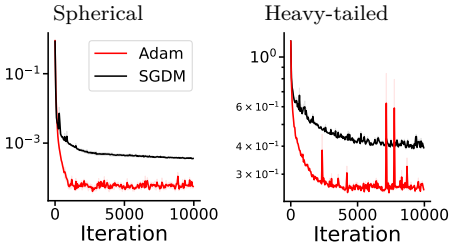
  $x \sim \mathrm{Unif}(\mathbb{S}^{d-1})$

- **Heavy-tailed covariates**

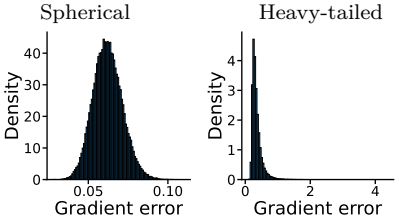
  $x \sim \sqrt{\Gamma_{0.1,10}} \cdot \mathrm{Unif}(\mathbb{S}^{d-1})$



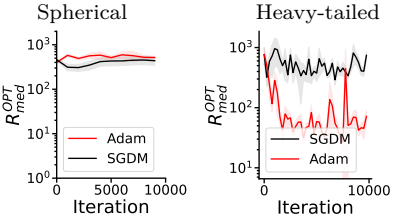
- **Comparing SGD v.s Adam:**

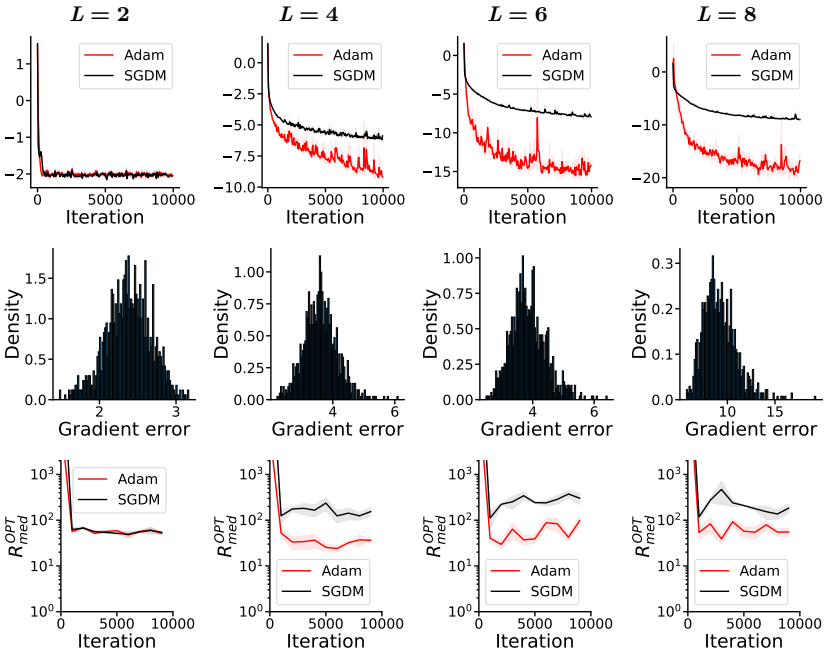


- **Stochastic Gradient Noise:**



- **Robust Condition Number:**




$L = 2$
$L = 4$
$L = 6$
$L = 8$
Adam
SGDM
Iteration
Density
Gradient error
R_{med}^{OPT}

Summary

The complexity of modern neural networks, especially transformers, often eludes precise mathematical understanding, and hence ***calls for such physics-style approaches based on simplified models****.*

We hope that our work will serve as the stepping stone for building a more precise theory of transformer optimization, as well as contributing to the development of efficient training methods for transformers.

- **Inspiration for ICL**
  - Simplification of Transformer Softmax $= \mathbb{I}$ or M

$$H = V \cdot \text{Softmax}\left(\frac{Q^\top K}{\sqrt{d}}\right) \implies H = \frac{1}{\sqrt{d}} X X^\top \widetilde{W} X \implies H = \frac{\widetilde{W} X}{\sqrt{d}}$$

$$\text{Attention} = PXM\left(X^\top QX\right) \qquad M \equiv \begin{bmatrix} \mathbb{I}_n & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{(n+1)\times(n+1)}$$

  - Multi-Layer Linear Transformer ($L \geq 3$)